

02 - Data Analysis

AI & Social Media Impact: Student Health & Grades

This notebook performs the main analysis of the cleaned student dataset.

Goals

- Understand the relationships between social media usage, AI usage, health, lifestyle, and academic performance.
- Compare important groups such as burnout levels and academic failure risk.
- Identify useful patterns and correlations.
- Summarize the main findings for later reporting and machine learning.

Input: data/processed/AI_SocialMedia_Student_Health_Dataset_clean.csv

1. Import Libraries

We import the libraries needed for data analysis and visualization.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Display more columns when showing DataFrames.
pd.set_option("display.max_columns", None)

# Make plots easier to read.
sns.set_theme(style="whitegrid")
```

2. Load the Cleaned Dataset

The cleaning step already created the processed CSV file. We use that file for the analysis.

```
In [2]: DATA_PATH = "../data/processed/AI_SocialMedia_Student_Health_Dataset_clean.csv"

df = pd.read_csv(DATA_PATH)

print(f"Dataset loaded successfully: {DATA_PATH}")
print(f"Rows: {df.shape[0]}")
print(f"Columns: {df.shape[1]}")
```

```
Dataset loaded successfully: ../data/processed/AI_SocialMedia_Student_Health_Data
set_clean.csv
Rows: 15000
Columns: 14
```

3. Quick Dataset Overview

First, we check the structure of the cleaned dataset before starting the detailed analysis.

```
In [3]: display(df.head())

print("\nData types:")
display(df.dtypes.to_frame("Data Type"))

print(f"\nMissing values: {df.isnull().sum().sum()}")
print(f"\nDuplicate rows: {df.duplicated().sum()}")
```

	Student_ID	Age	Gender	Education_Level	Daily_Social_Media_Hours	Daily_AI_Tool_Usa
0	STU_00001	19	Male	College	5.32	
1	STU_00002	16	Non-binary	High School	5.21	
2	STU_00003	25	Male	University	3.61	
3	STU_00004	23	Male	University	2.93	
4	STU_00005	20	Non-binary	College	6.52	

Data types:

	Data Type
Student_ID	str
Age	int64
Gender	str
Education_Level	str
Daily_Social_Media_Hours	float64
Daily_AI_Tool_Usage_Hours	float64
Sleep_Hours	float64
Physical_Activity_Hours	float64
Mental_Health_Score	float64
Physical_Health_Score	float64
Social_Isolation_Score	float64
Burnout_Level	str
Academic_Performance_Score	float64
Academic_Failure_Risk	int64

Missing values: 0
Duplicate rows: 0

4. Separate Numerical and Categorical Columns

Separating the columns makes it easier to perform the correct type of analysis.

```
In [4]: numerical_columns = df.select_dtypes(include=np.number).columns.tolist()
categorical_columns = df.select_dtypes(include=["object", "category"]).columns.t

print("Numerical columns:")
for column in numerical_columns:
    print(f"- {column}")

print("\nCategorical columns:")
for column in categorical_columns:
    print(f"- {column}")
```

Numerical columns:

- Age
- Daily_Social_Media_Hours
- Daily_AI_Tool_Usage_Hours
- Sleep_Hours
- Physical_Activity_Hours
- Mental_Health_Score
- Physical_Health_Score
- Social_Isolation_Score
- Academic_Performance_Score
- Academic_Failure_Risk

Categorical columns:

- Student_ID
- Gender
- Education_Level
- Burnout_Level

C:\Users\ayman\AppData\Local\Temp\ipykernel_17644\3255864776.py:2: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.

See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
categorical_columns = df.select_dtypes(include=["object", "category"]).columns.tolist()
```

5. Descriptive Statistics

These statistics help us understand the typical values and variation of the numerical variables.

```
In [5]: summary = df[numerical_columns].describe().T
summary["median"] = df[numerical_columns].median()
summary["variance"] = df[numerical_columns].var()
summary["IQR"] = df[numerical_columns].quantile(0.75) - df[numerical_columns].qu

display(summary.round(2))
```

	count	mean	std	min	25%	50%	75%	max	me
Age	15000.0	19.04	3.77	13.00	16.00	19.00	22.00	25.00	1
Daily_Social_Media_Hours	15000.0	4.54	2.40	0.00	2.83	4.50	6.18	14.00	
Daily_AI_Tool_Usage_Hours	15000.0	2.58	1.68	0.00	1.31	2.51	3.74	9.50	
Sleep_Hours	15000.0	6.54	1.27	2.00	5.68	6.55	7.41	11.15	
Physical_Activity_Hours	15000.0	1.25	1.04	0.00	0.31	1.13	1.95	5.00	
Mental_Health_Score	15000.0	72.51	9.25	32.56	66.78	73.76	79.58	91.76	7
Physical_Health_Score	15000.0	88.02	9.68	48.03	81.52	89.47	97.16	99.98	8
Social_Isolation_Score	15000.0	4.31	1.16	0.97	3.47	4.28	5.12	8.41	
Academic_Performance_Score	15000.0	78.31	11.54	23.99	70.53	78.59	86.70	99.98	7
Academic_Failure_Risk	15000.0	0.06	0.24	0.00	0.00	0.00	0.00	1.00	

6. Categorical Variables

We examine the number and percentage of students in each category.

```
In [6]: for column in ["Gender", "Education_Level", "Burnout_Level"]:
        print("=" * 70)
        print(column)
        print("=" * 70)

        counts = df[column].value_counts()
        percentages = df[column].value_counts(normalize=True).mul(100)

        result = pd.DataFrame({
            "Count": counts,
            "Percentage": percentages.round(2)
        })

        display(result)
```

```
=====
Gender
=====
```

	Count	Percentage
Gender		
Male	7265	48.43
Female	7143	47.62
Non-binary	592	3.95

```
=====
Education_Level
=====
```

	Count	Percentage
Education_Level		
High School	5693	37.95
University	4710	31.40
College	4597	30.65

=====

Burnout_Level

=====

	Count	Percentage
Burnout_Level		
Low	9000	60.0
Moderate	4200	28.0
High	1350	9.0
Severe	450	3.0

7. Target Variable: Academic Failure Risk

`Academic_Failure_Risk` is the main target variable for the project.

- `0` = No academic failure risk
- `1` = Academic failure risk

We first check how balanced the two classes are.

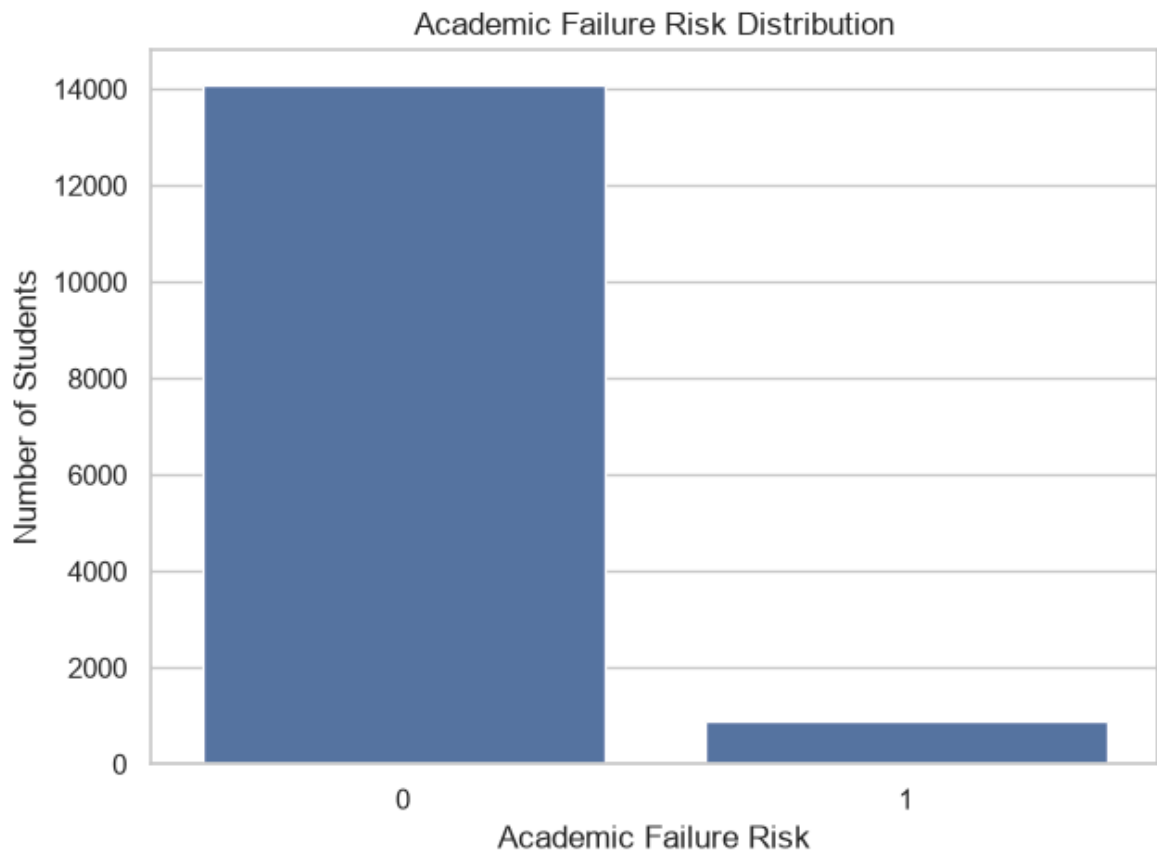
```
In [7]: target_counts = df["Academic_Failure_Risk"].value_counts().sort_index()
target_percentages = df["Academic_Failure_Risk"].value_counts(normalize=True).so

target_summary = pd.DataFrame({
    "Count": target_counts,
    "Percentage": target_percentages.round(2)
})

display(target_summary)

plt.figure(figsize=(7, 5))
sns.countplot(data=df, x="Academic_Failure_Risk")
plt.title("Academic Failure Risk Distribution")
plt.xlabel("Academic Failure Risk")
plt.ylabel("Number of Students")
plt.show()
```

	Count	Percentage
Academic_Failure_Risk		
0	14100	94.0
1	900	6.0



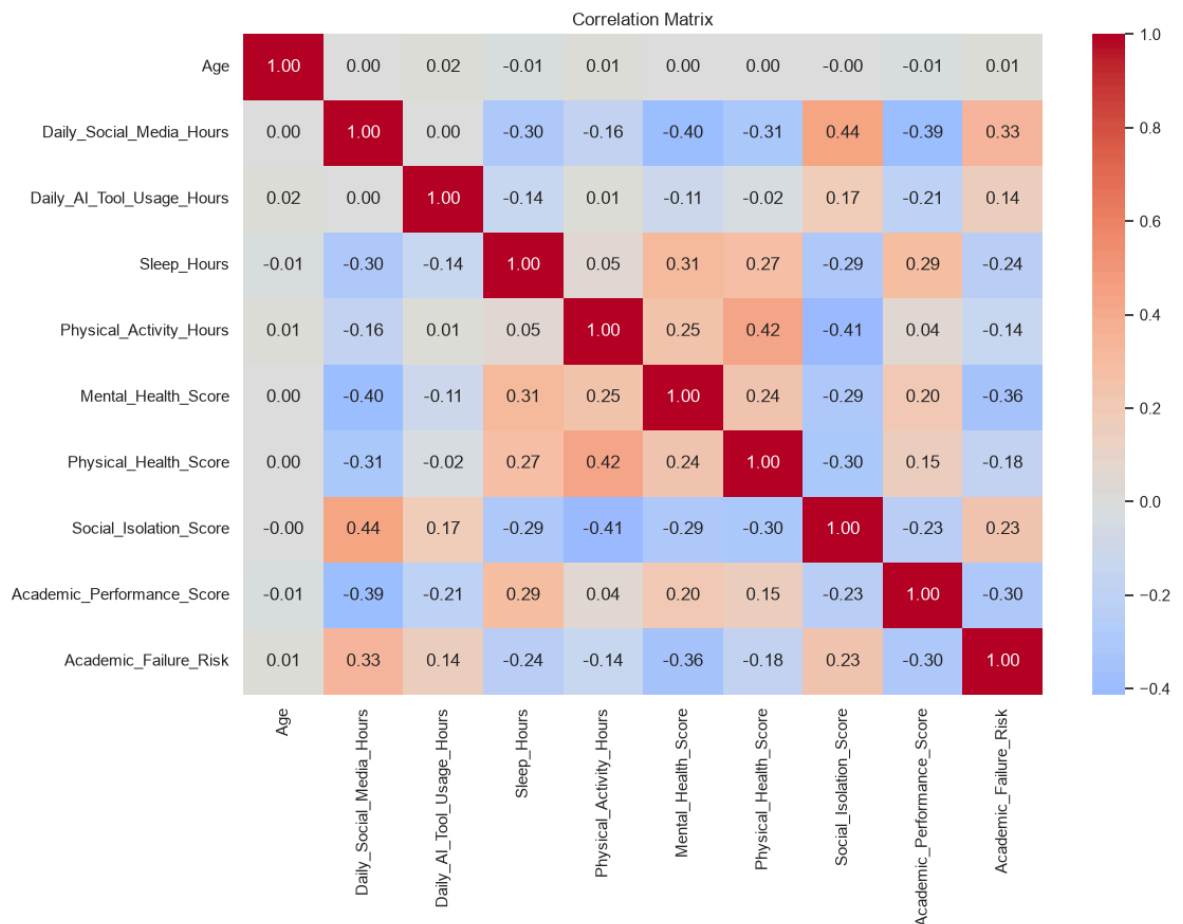
8. Correlation Analysis

Correlation shows how strongly two numerical variables move together.

- Positive correlation: both variables tend to increase together.
- Negative correlation: one variable tends to increase while the other decreases.
- Values close to 0 indicate a weak linear relationship.

```
In [8]: correlation_matrix = df[numerical_columns].corr()

plt.figure(figsize=(12, 9))
sns.heatmap(
    correlation_matrix,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0
)
plt.title("Correlation Matrix")
plt.tight_layout()
plt.show()
```



9. Strongest Relationships with Academic Performance

Academic performance is one of the most important outcomes in this dataset. We rank the correlations with `Academic_Performance_Score`.

```
In [9]: academic_correlations = (
    correlation_matrix["Academic_Performance_Score"]
    .drop("Academic_Performance_Score")
    .sort_values(key=abs, ascending=False)
)

display(academic_correlations.to_frame("Correlation with Academic Performance").
```

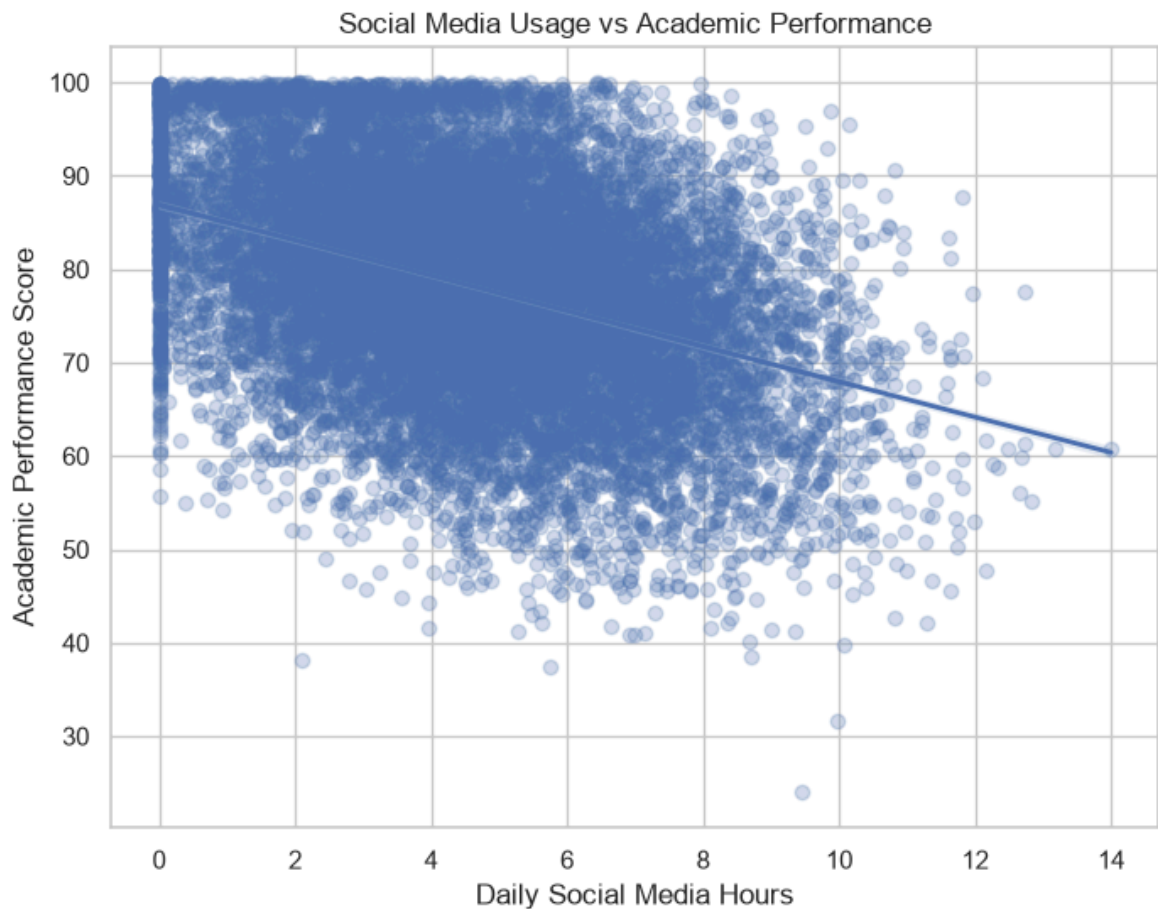
Correlation with Academic Performance	
Daily_Social_Media_Hours	-0.394
Academic_Failure_Risk	-0.296
Sleep_Hours	0.292
Social_Isolation_Score	-0.235
Daily_AI_Tool_Usage_Hours	-0.205
Mental_Health_Score	0.204
Physical_Health_Score	0.151
Physical_Activity_Hours	0.042
Age	-0.008

10. Social Media Usage and Academic Performance

We investigate whether daily social media usage is associated with academic performance.

```
In [10]: plt.figure(figsize=(8, 6))
sns.regplot(
    data=df,
    x="Daily_Social_Media_Hours",
    y="Academic_Performance_Score",
    scatter_kws={"alpha": 0.25},
    line_kws={"linewidth": 2}
)
plt.title("Social Media Usage vs Academic Performance")
plt.xlabel("Daily Social Media Hours")
plt.ylabel("Academic Performance Score")
plt.show()

social_corr = df["Daily_Social_Media_Hours"].corr(df["Academic_Performance_Score"])
print(f"Correlation: {social_corr:.3f}")
```



Correlation: -0.394

11. Social Media Usage and Academic Failure Risk

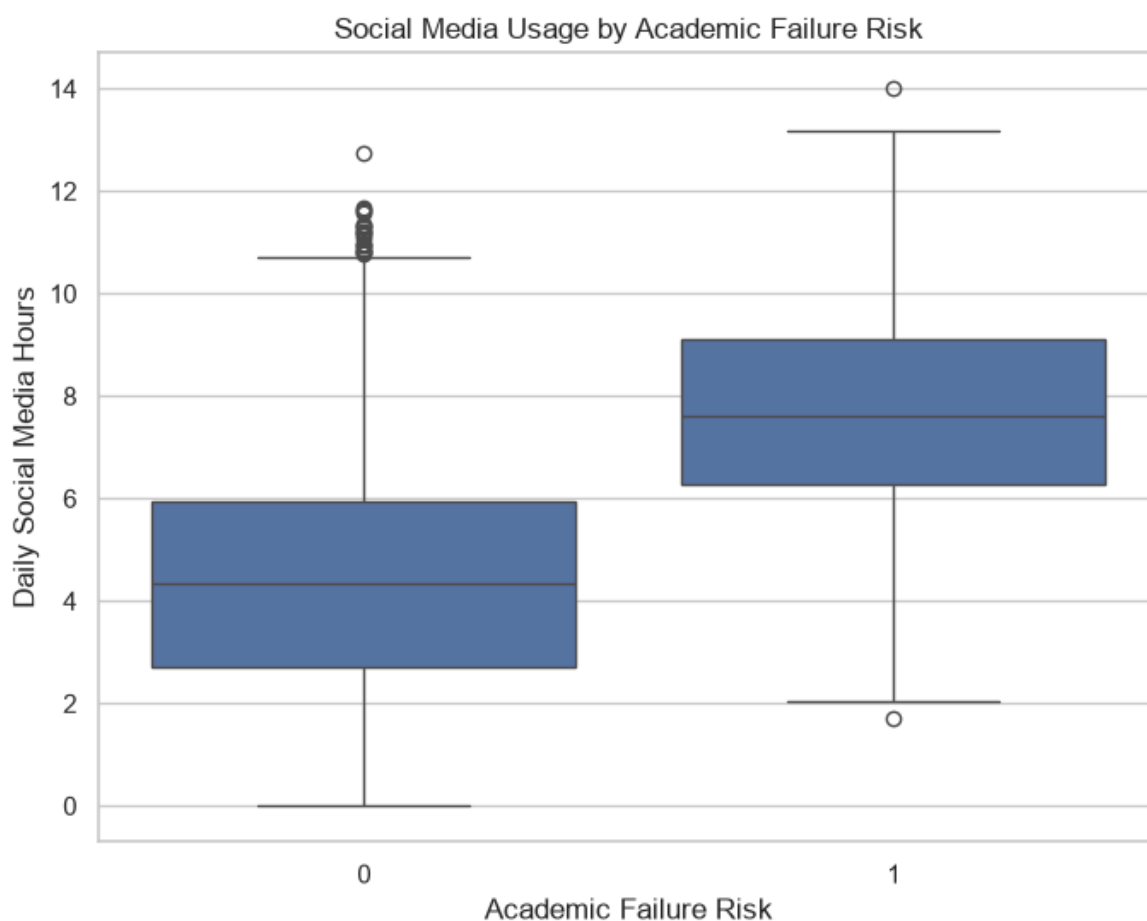
Now we compare social media usage between students with and without academic failure risk.

```
In [11]: social_risk_summary = df.groupby("Academic_Failure_Risk")["Daily_Social_Media_Hours"]
        .agg(["count", "mean", "median", "min", "max"])

        display(social_risk_summary.round(2))

        plt.figure(figsize=(8, 6))
        sns.boxplot(
            data=df,
            x="Academic_Failure_Risk",
            y="Daily_Social_Media_Hours"
        )
        plt.title("Social Media Usage by Academic Failure Risk")
        plt.xlabel("Academic Failure Risk")
        plt.ylabel("Daily Social Media Hours")
        plt.show()
```

	count	mean	median	min	max
Academic_Failure_Risk					
0	14100	4.34	4.33	0.0	12.73
1	900	7.67	7.60	1.7	14.00



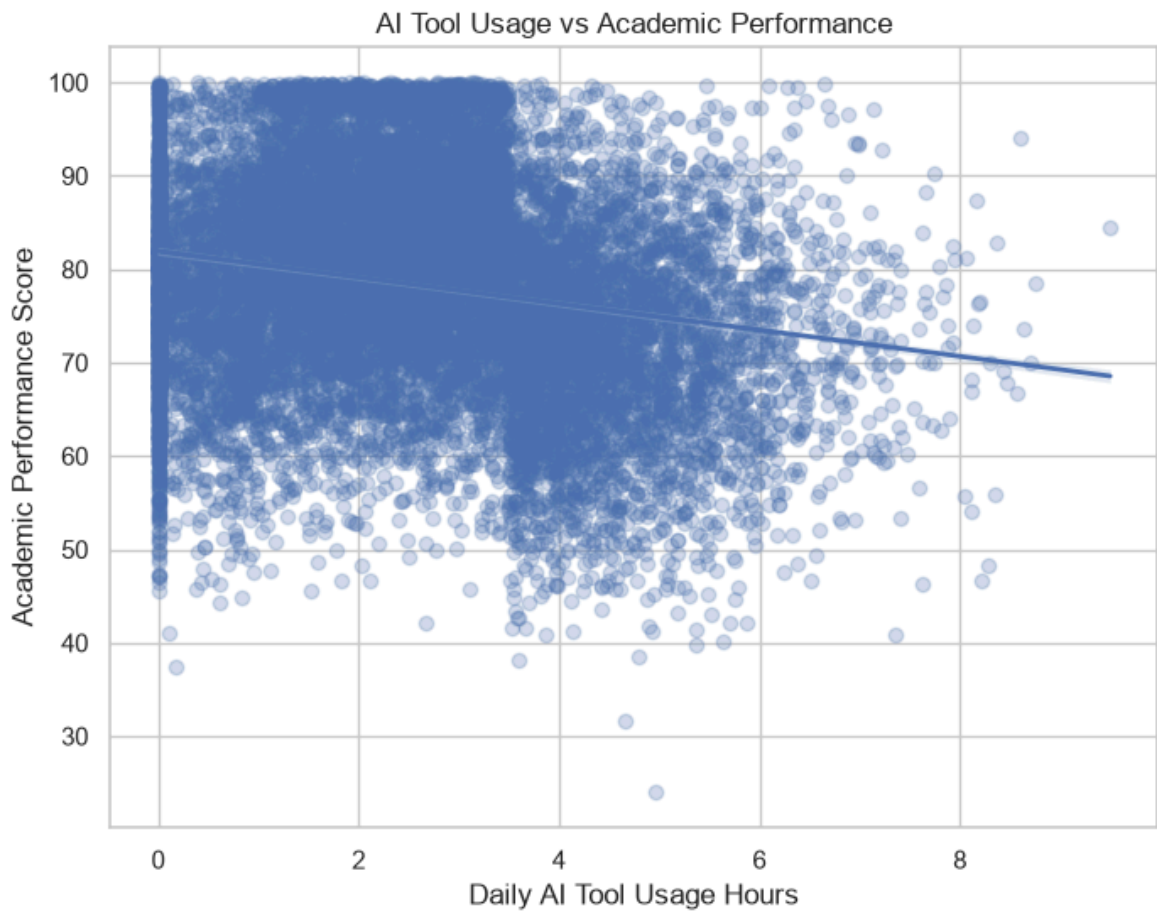
12. AI Tool Usage and Academic Performance

We examine the relationship between daily AI tool usage and academic performance.

```
In [12]: ai_corr = df["Daily_AI_Tool_Usage_Hours"].corr(df["Academic_Performance_Score"])
print(f"Correlation between AI usage and academic performance: {ai_corr:.3f}")

plt.figure(figsize=(8, 6))
sns.regplot(
    data=df,
    x="Daily_AI_Tool_Usage_Hours",
    y="Academic_Performance_Score",
    scatter_kws={"alpha": 0.25},
    line_kws={"linewidth": 2}
)
plt.title("AI Tool Usage vs Academic Performance")
plt.xlabel("Daily AI Tool Usage Hours")
plt.ylabel("Academic Performance Score")
plt.show()
```

Correlation between AI usage and academic performance: -0.205



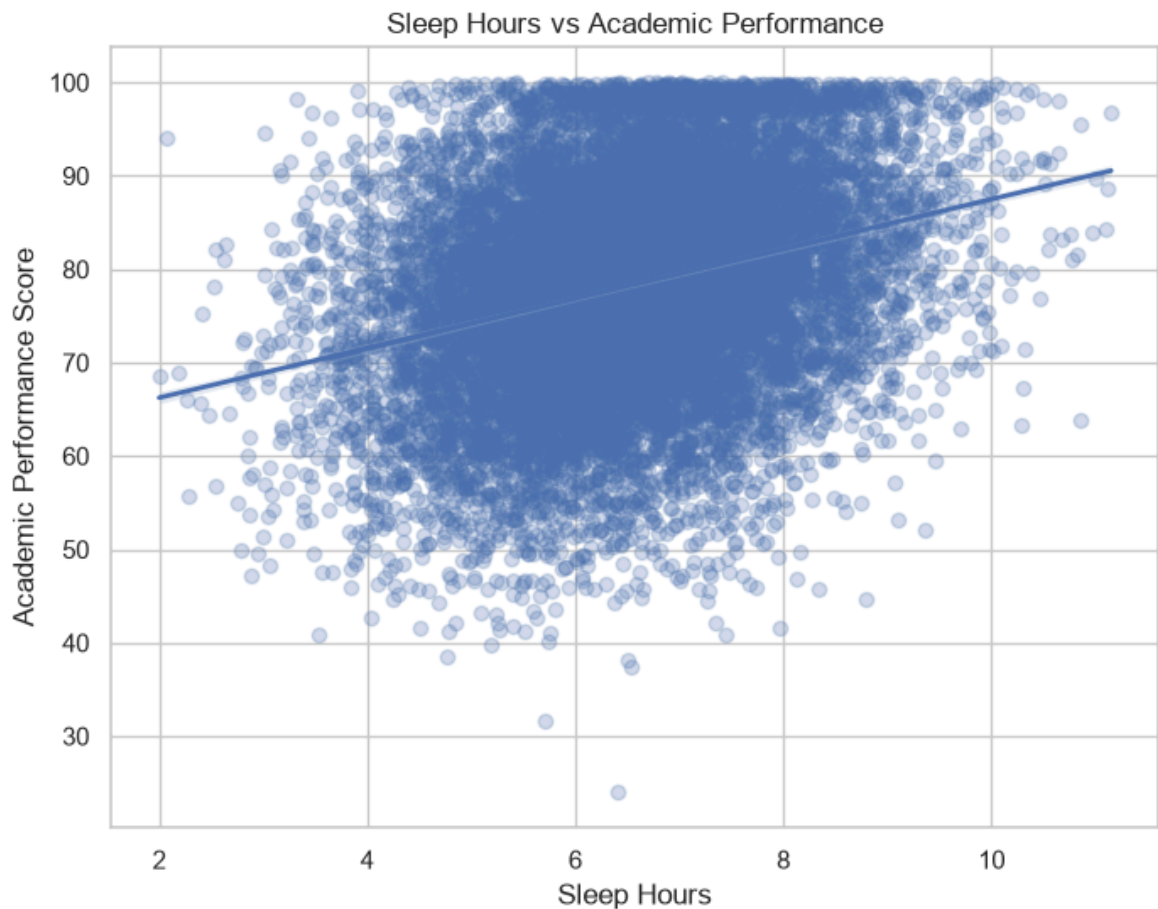
13. Sleep and Academic Performance

Sleep is an important lifestyle factor. We check its relationship with academic performance.

```
In [13]: sleep_corr = df["Sleep_Hours"].corr(df["Academic_Performance_Score"])
print(f"Correlation between sleep and academic performance: {sleep_corr:.3f}")

plt.figure(figsize=(8, 6))
sns.regplot(
    data=df,
    x="Sleep_Hours",
    y="Academic_Performance_Score",
    scatter_kws={"alpha": 0.25},
    line_kws={"linewidth": 2}
)
plt.title("Sleep Hours vs Academic Performance")
plt.xlabel("Sleep Hours")
plt.ylabel("Academic Performance Score")
plt.show()
```

Correlation between sleep and academic performance: 0.292



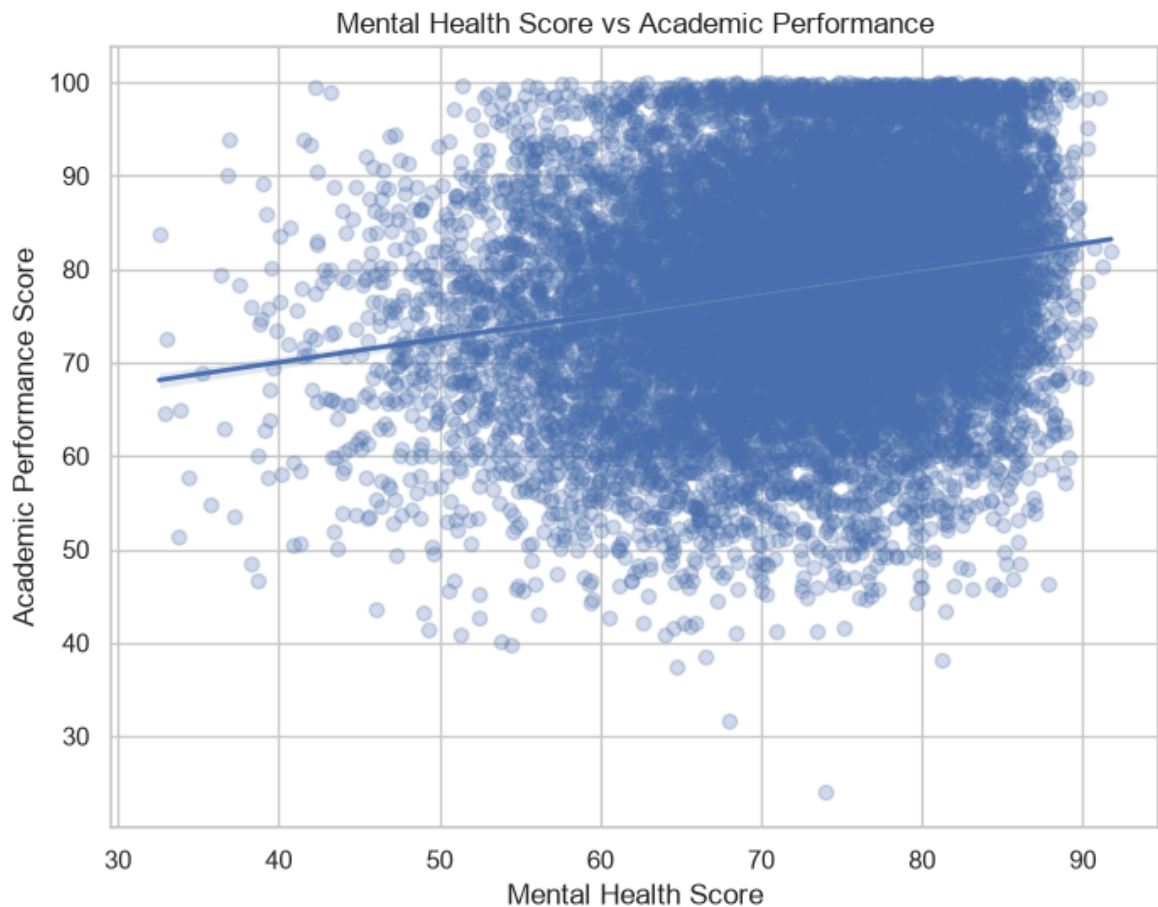
14. Mental Health and Academic Performance

We investigate whether mental health scores are associated with academic performance.

```
In [14]: mental_corr = df["Mental_Health_Score"].corr(df["Academic_Performance_Score"])
print(f"Correlation between mental health and academic performance: {mental_corr}")

plt.figure(figsize=(8, 6))
sns.regplot(
    data=df,
    x="Mental_Health_Score",
    y="Academic_Performance_Score",
    scatter_kws={"alpha": 0.25},
    line_kws={"linewidth": 2}
)
plt.title("Mental Health Score vs Academic Performance")
plt.xlabel("Mental Health Score")
plt.ylabel("Academic Performance Score")
plt.show()
```

Correlation between mental health and academic performance: 0.204



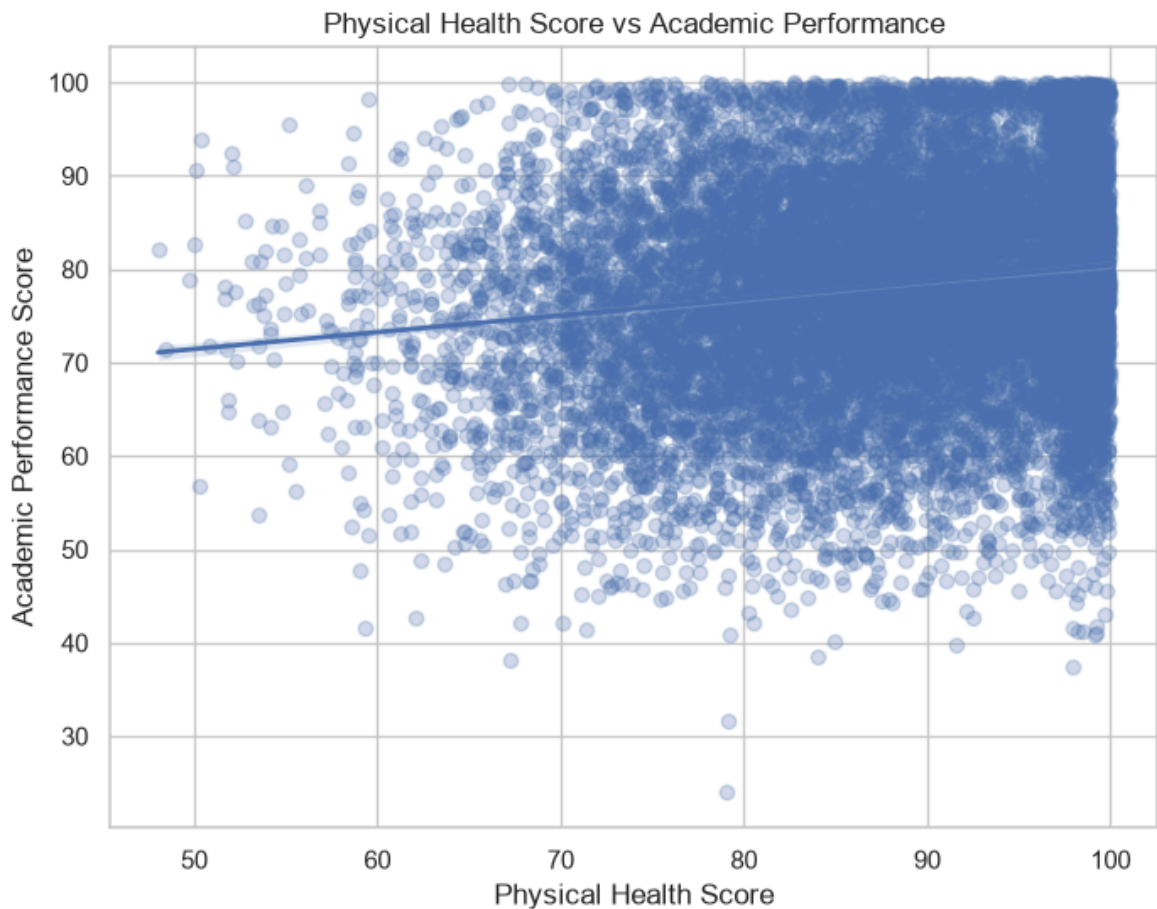
15. Physical Health and Academic Performance

We compare physical health scores with academic performance.

```
In [15]: physical_corr = df["Physical_Health_Score"].corr(df["Academic_Performance_Score"])
print(f"Correlation between physical health and academic performance: {physical_corr}")

plt.figure(figsize=(8, 6))
sns.regplot(
    data=df,
    x="Physical_Health_Score",
    y="Academic_Performance_Score",
    scatter_kws={"alpha": 0.25},
    line_kws={"linewidth": 2}
)
plt.title("Physical Health Score vs Academic Performance")
plt.xlabel("Physical Health Score")
plt.ylabel("Academic Performance Score")
plt.show()
```

Correlation between physical health and academic performance: 0.151



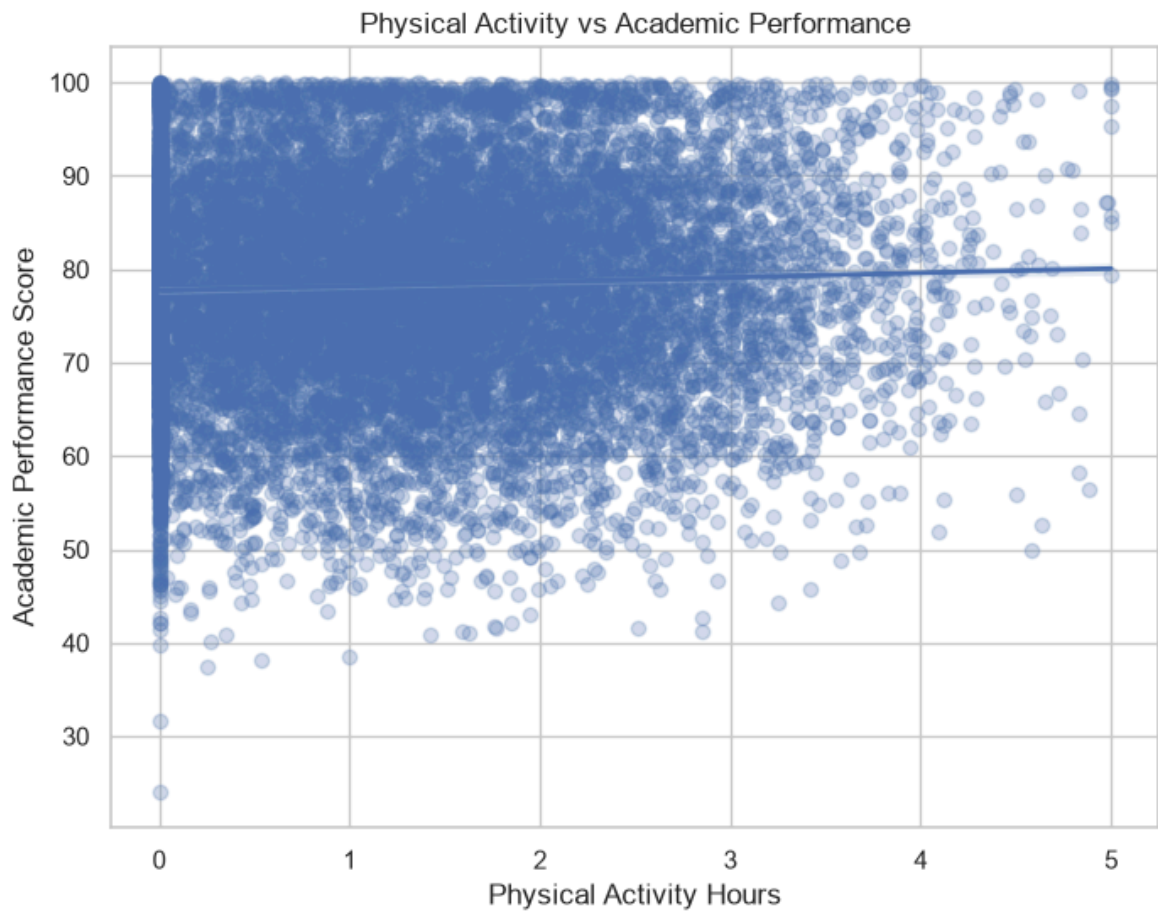
16. Physical Activity and Academic Performance

We check whether physical activity has a noticeable relationship with academic performance.

```
In [16]: activity_corr = df["Physical_Activity_Hours"].corr(df["Academic_Performance_Score"])
print(f"Correlation between physical activity and academic performance: {activity_corr}")

plt.figure(figsize=(8, 6))
sns.regplot(
    data=df,
    x="Physical_Activity_Hours",
    y="Academic_Performance_Score",
    scatter_kws={"alpha": 0.25},
    line_kws={"linewidth": 2}
)
plt.title("Physical Activity vs Academic Performance")
plt.xlabel("Physical Activity Hours")
plt.ylabel("Academic Performance Score")
plt.show()
```

Correlation between physical activity and academic performance: 0.042



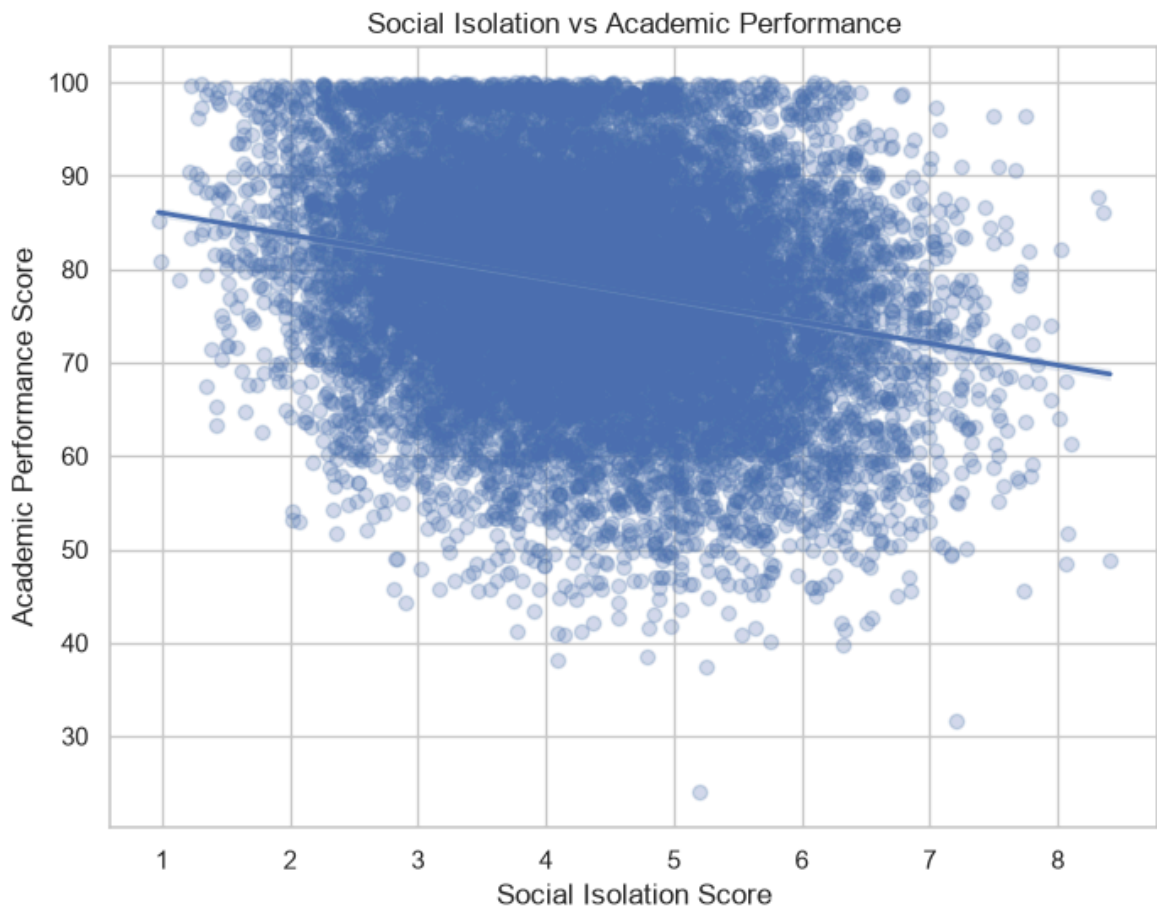
17. Social Isolation and Academic Performance

We examine whether social isolation is associated with academic performance.

```
In [17]: isolation_corr = df["Social_Isolation_Score"].corr(df["Academic_Performance_Score"])
print(f"Correlation between social isolation and academic performance: {isolation_corr}")

plt.figure(figsize=(8, 6))
sns.regplot(
    data=df,
    x="Social_Isolation_Score",
    y="Academic_Performance_Score",
    scatter_kws={"alpha": 0.25},
    line_kws={"linewidth": 2}
)
plt.title("Social Isolation vs Academic Performance")
plt.xlabel("Social Isolation Score")
plt.ylabel("Academic Performance Score")
plt.show()
```

Correlation between social isolation and academic performance: -0.235



18. Burnout Level and Academic Performance

Burnout is divided into four levels: Low, Moderate, High, and Severe. We compare academic performance across these groups.

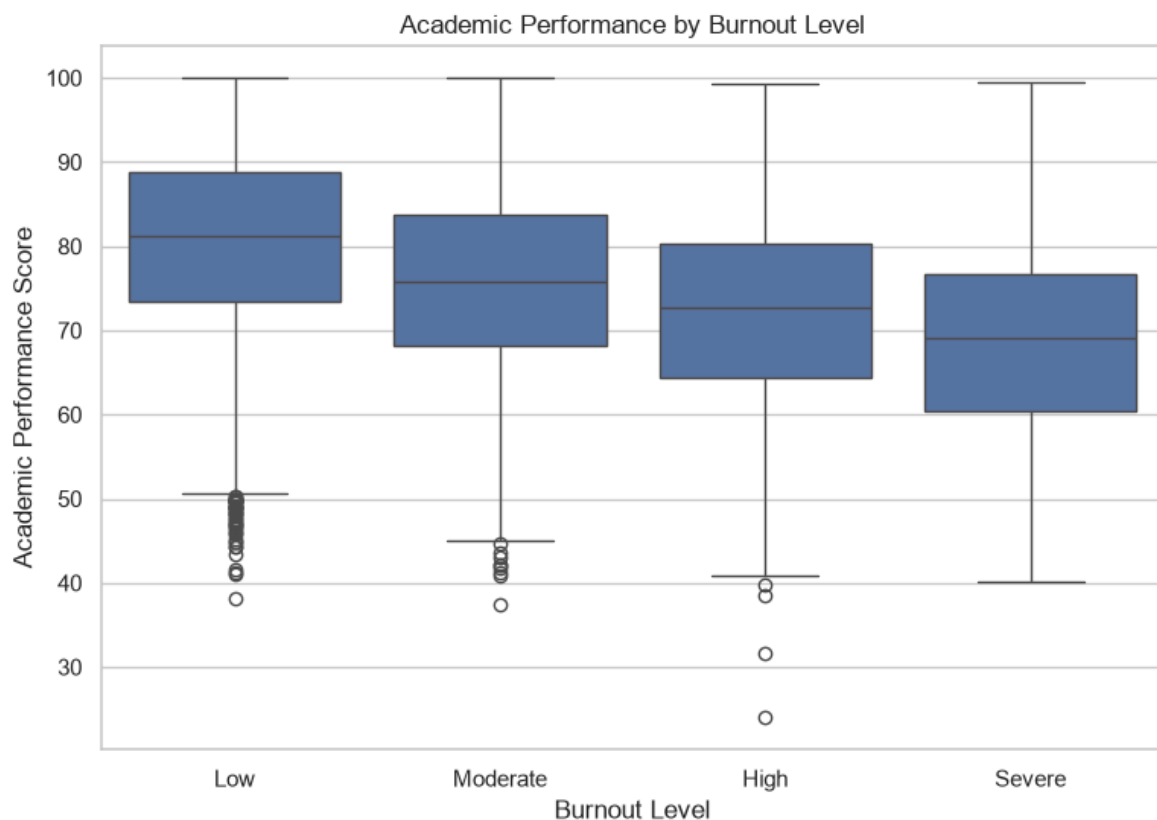
```
In [18]: burnout_order = ["Low", "Moderate", "High", "Severe"]

burnout_performance = df.groupby("Burnout_Level")[
    "Academic_Performance_Score"
].agg(["count", "mean", "median", "std", "min", "max"]).reindex(burnout_order)

display(burnout_performance.round(2))

plt.figure(figsize=(9, 6))
sns.boxplot(
    data=df,
    x="Burnout_Level",
    y="Academic_Performance_Score",
    order=burnout_order
)
plt.title("Academic Performance by Burnout Level")
plt.xlabel("Burnout Level")
plt.ylabel("Academic Performance Score")
plt.show()
```

	count	mean	median	std	min	max
Burnout_Level						
Low	9000	80.86	81.26	10.88	38.19	99.97
Moderate	4200	75.79	75.77	11.22	37.36	99.98
High	1350	72.44	72.63	11.31	23.99	99.34
Severe	450	68.64	69.04	11.55	40.08	99.48



19. Burnout Level and Academic Failure Risk

We calculate the percentage of students at academic failure risk within each burnout group.

```
In [19]: burnout_risk = pd.crosstab(
    df["Burnout_Level"],
    df["Academic_Failure_Risk"],
    normalize="index"
).mul(100).reindex(burnout_order)

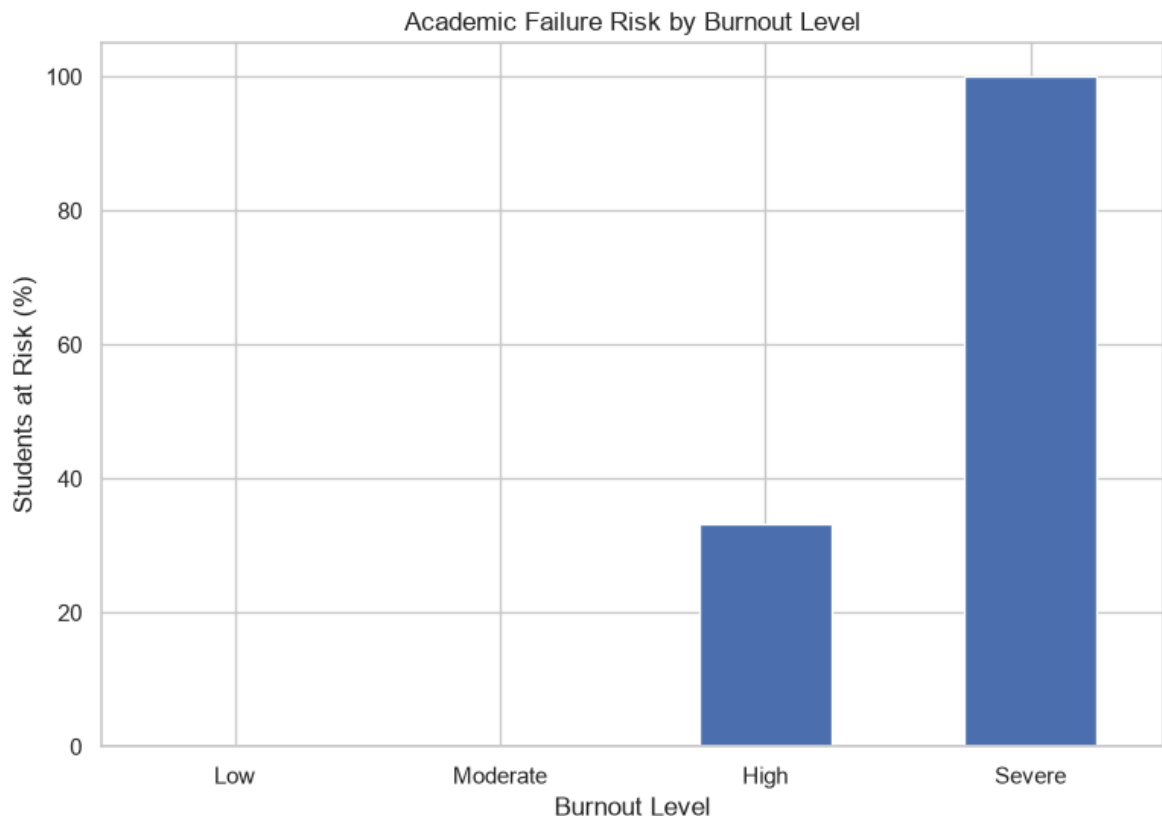
burnout_risk.columns = ["No Risk (%)", "Failure Risk (%)"]

display(burnout_risk.round(2))

burnout_risk["Failure Risk (%)"].plot(kind="bar", figsize=(9, 6))
plt.title("Academic Failure Risk by Burnout Level")
plt.xlabel("Burnout Level")
plt.ylabel("Students at Risk (%)")
```

```
plt.xticks(rotation=0)
plt.show()
```

	No Risk (%)	Failure Risk (%)
Burnout_Level		
Low	100.00	0.00
Moderate	100.00	0.00
High	66.67	33.33
Severe	0.00	100.00



20. Gender and Academic Performance

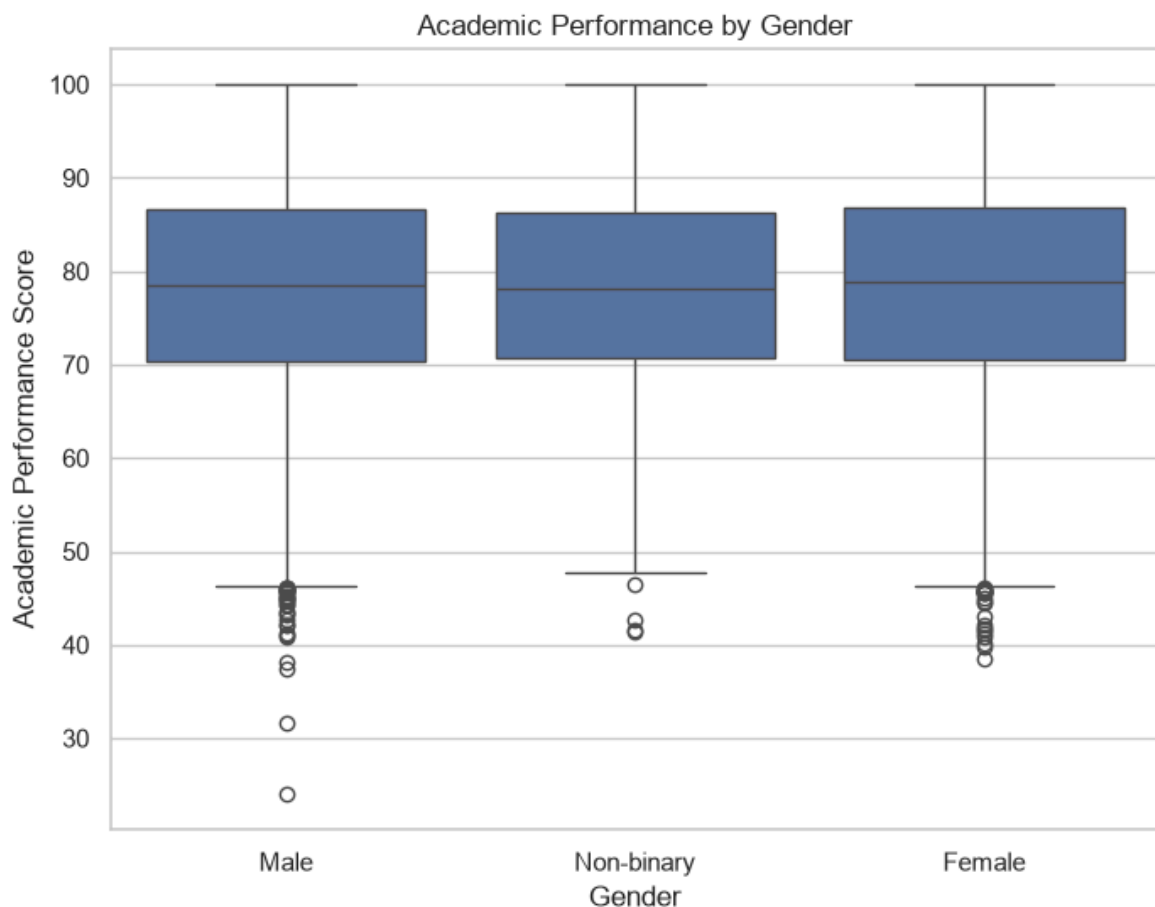
We compare average academic performance across gender groups.

```
In [20]: gender_analysis = df.groupby("Gender")[
        "Academic_Performance_Score"
    ].agg(["count", "mean", "median", "std"]).sort_values("mean", ascending=False)

display(gender_analysis.round(2))

plt.figure(figsize=(8, 6))
sns.boxplot(data=df, x="Gender", y="Academic_Performance_Score")
plt.title("Academic Performance by Gender")
plt.xlabel("Gender")
plt.ylabel("Academic Performance Score")
plt.show()
```

	count	mean	median	std
Gender				
Female	7143	78.45	78.77	11.60
Male	7265	78.20	78.46	11.49
Non-binary	592	78.07	78.17	11.56



21. Education Level and Academic Performance

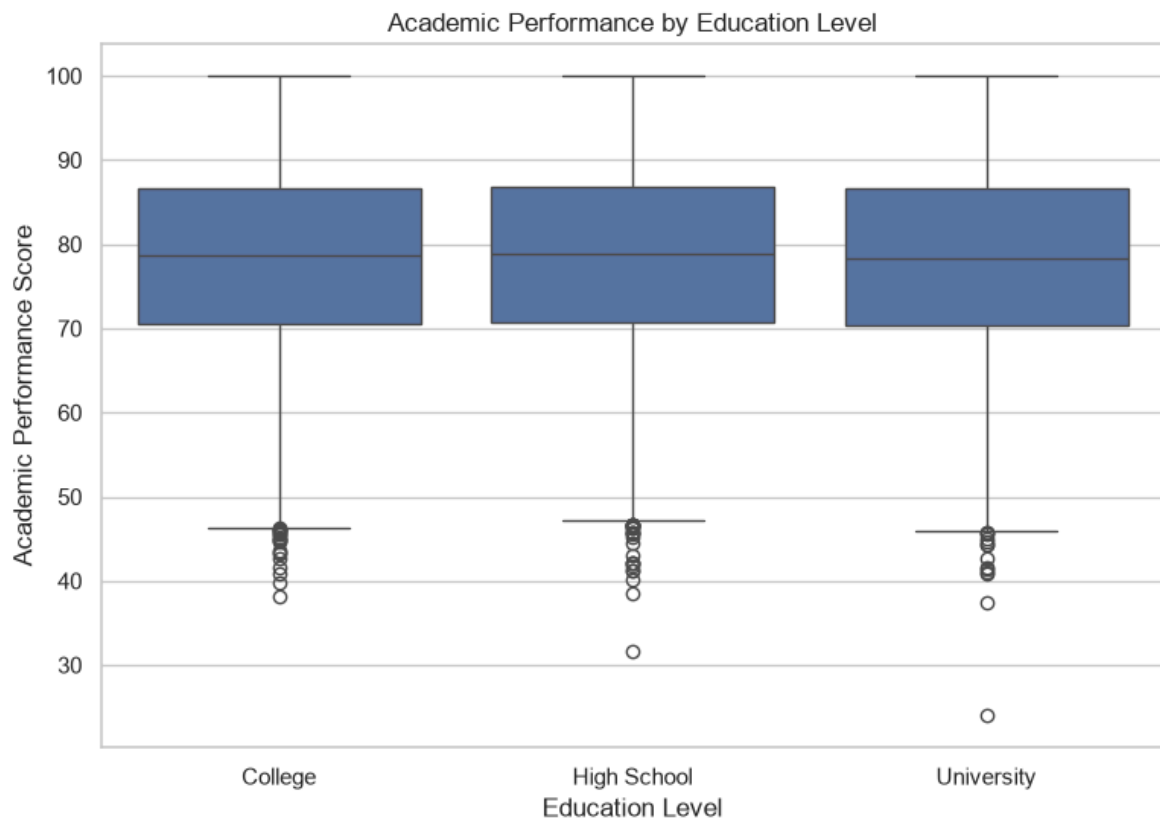
We compare academic performance among High School, College, and University students.

```
In [21]: education_analysis = df.groupby("Education_Level")["Academic_Performance_Score"]
        .agg(["count", "mean", "median", "std"]).sort_values("mean", ascending=False)

        display(education_analysis.round(2))

        plt.figure(figsize=(9, 6))
        sns.boxplot(data=df, x="Education_Level", y="Academic_Performance_Score")
        plt.title("Academic Performance by Education Level")
        plt.xlabel("Education Level")
        plt.ylabel("Academic Performance Score")
        plt.show()
```

	count	mean	median	std
Education_Level				
High School	5693	78.46	78.80	11.46
College	4597	78.31	78.70	11.57
University	4710	78.14	78.26	11.62



22. Group Analysis: Social Media Usage and Burnout

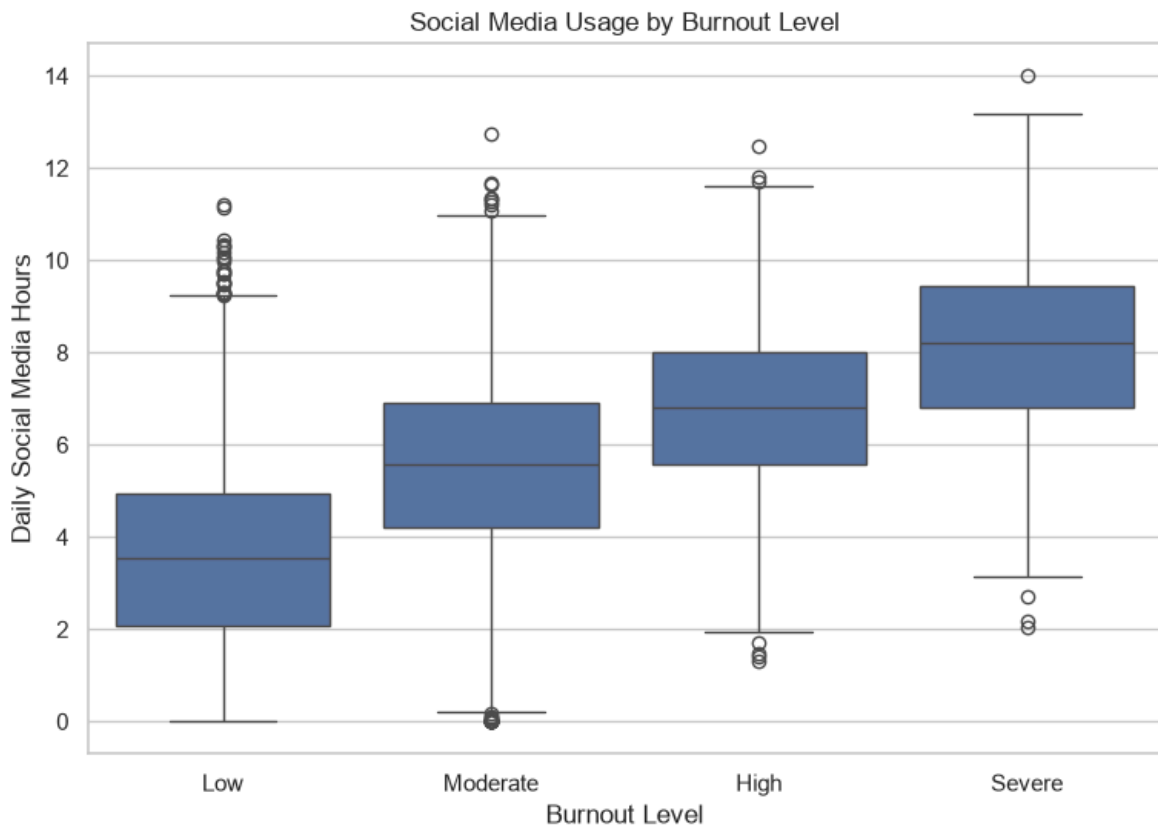
We compare average social media usage across burnout levels.

```
In [22]: burnout_social = df.groupby("Burnout_Level")[
    "Daily_Social_Media_Hours"
].agg(["count", "mean", "median", "std"]).reindex(burnout_order)

display(burnout_social.round(2))

plt.figure(figsize=(9, 6))
sns.boxplot(
    data=df,
    x="Burnout_Level",
    y="Daily_Social_Media_Hours",
    order=burnout_order
)
plt.title("Social Media Usage by Burnout Level")
plt.xlabel("Burnout Level")
plt.ylabel("Daily Social Media Hours")
plt.show()
```

	count	mean	median	std
Burnout_Level				
Low	9000	3.55	3.53	2.04
Moderate	4200	5.54	5.56	2.00
High	1350	6.80	6.81	1.84
Severe	450	8.14	8.22	1.95



23. Group Analysis: Sleep and Burnout

We compare average sleep hours across burnout levels.

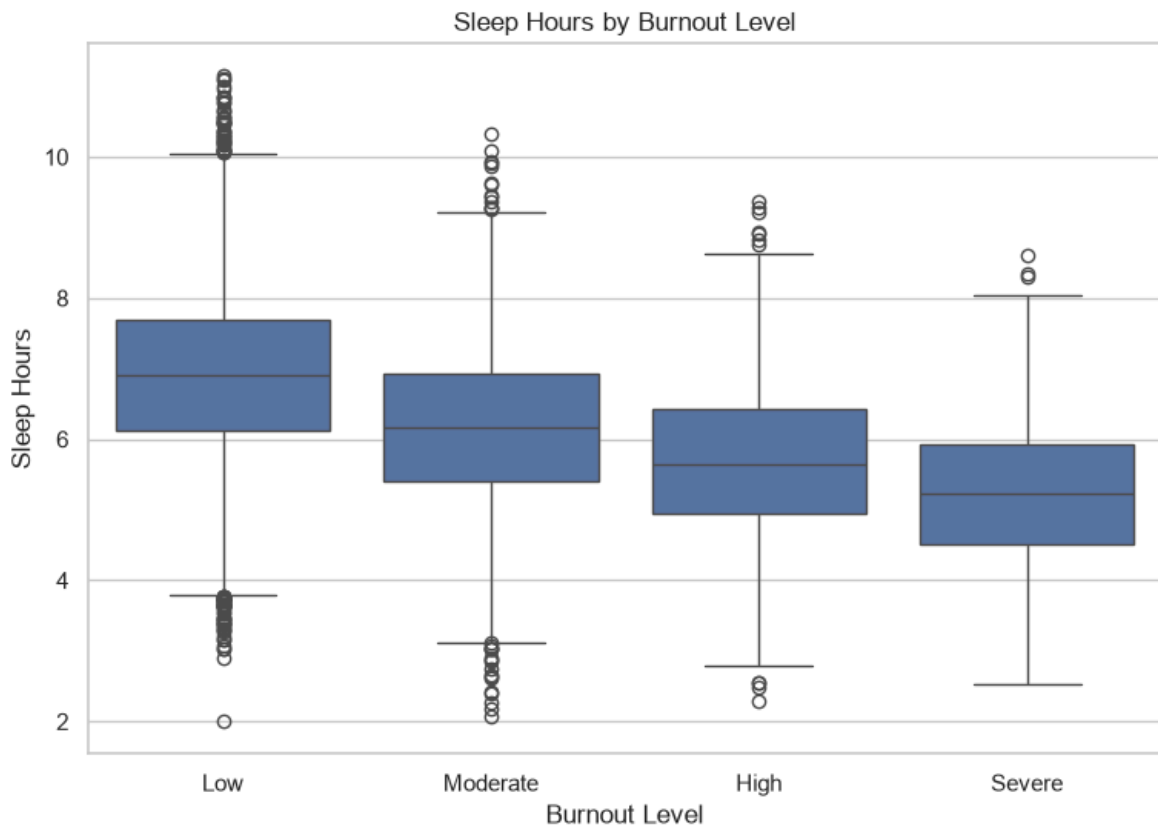
```
In [23]: burnout_sleep = df.groupby("Burnout_Level")["Sleep_Hours"]
        .agg(["count", "mean", "median", "std"]).reindex(burnout_order)

        display(burnout_sleep.round(2))

        plt.figure(figsize=(9, 6))
        sns.boxplot(
            data=df,
            x="Burnout_Level",
            y="Sleep_Hours",
            order=burnout_order
        )
        plt.title("Sleep Hours by Burnout Level")
        plt.xlabel("Burnout Level")
```

```
plt.ylabel("Sleep Hours")
plt.show()
```

	count	mean	median	std
Burnout_Level				
Low	9000	6.91	6.90	1.19
Moderate	4200	6.17	6.18	1.14
High	1350	5.67	5.65	1.13
Severe	450	5.24	5.24	1.07



24. Group Analysis: Mental Health and Burnout

We compare mental health scores across burnout groups.

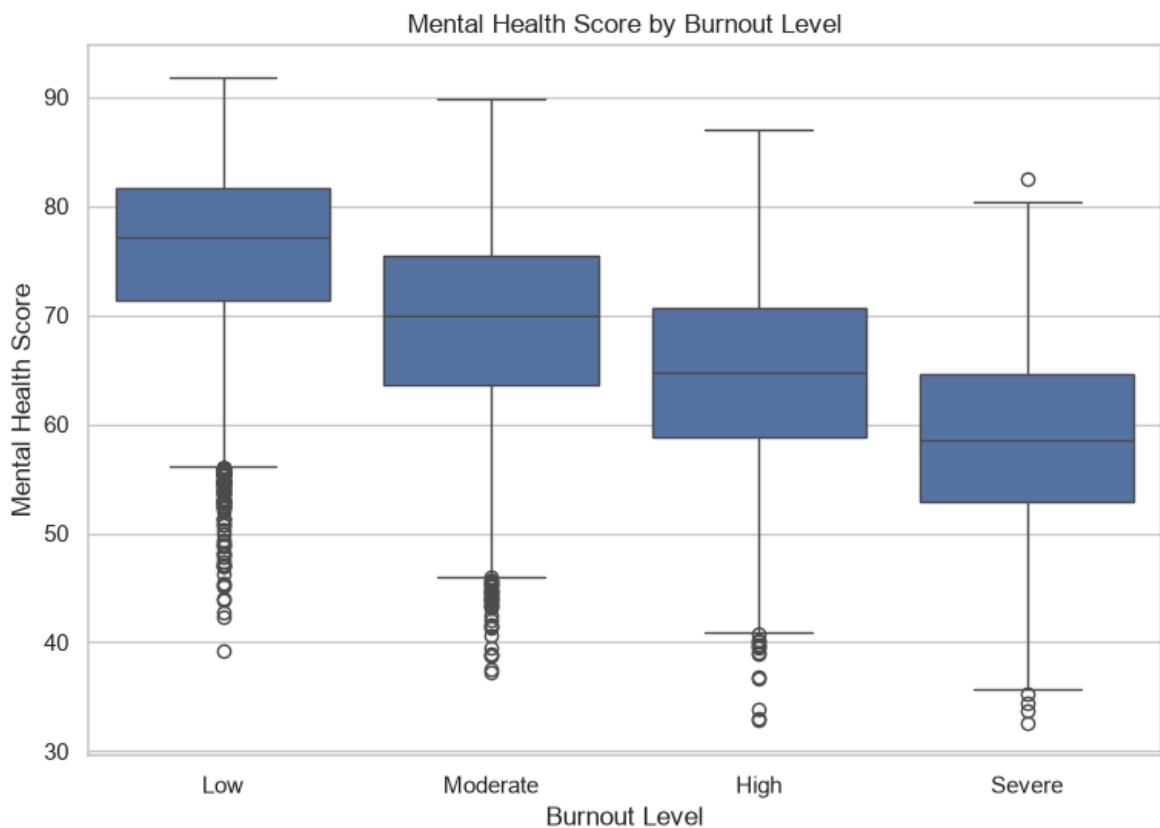
```
In [24]: burnout_mental = df.groupby("Burnout_Level")[
        "Mental_Health_Score"
    ].agg(["count", "mean", "median", "std"]).reindex(burnout_order)

display(burnout_mental.round(2))

plt.figure(figsize=(9, 6))
sns.boxplot(
    data=df,
    x="Burnout_Level",
    y="Mental_Health_Score",
    order=burnout_order
)
plt.title("Mental Health Score by Burnout Level")
```

```
plt.xlabel("Burnout Level")
plt.ylabel("Mental Health Score")
plt.show()
```

	count	mean	median	std
Burnout_Level				
Low	9000	76.01	77.09	7.41
Moderate	4200	69.16	69.94	8.50
High	1350	64.27	64.72	9.17
Severe	450	58.62	58.59	8.96



25. Factors Associated with Academic Failure Risk

This table ranks numerical variables by their correlation with the target variable

`Academic_Failure_Risk` .

```
In [25]: risk_correlations = (
    correlation_matrix["Academic_Failure_Risk"]
    .drop("Academic_Failure_Risk")
    .sort_values(key=abs, ascending=False)
)

display(
    risk_correlations
    .to_frame("Correlation with Failure Risk")
    .round(3)
)
```

Correlation with Failure Risk	
Mental_Health_Score	-0.356
Daily_Social_Media_Hours	0.330
Academic_Performance_Score	-0.296
Sleep_Hours	-0.238
Social_Isolation_Score	0.231
Physical_Health_Score	-0.182
Daily_AI_Tool_Usage_Hours	0.139
Physical_Activity_Hours	-0.137
Age	0.011

26. Compare Students With and Without Failure Risk

Here we compare the average values of important variables for the two target groups.

```
In [26]: important_columns = [
    "Daily_Social_Media_Hours",
    "Daily_AI_Tool_Usage_Hours",
    "Sleep_Hours",
    "Physical_Activity_Hours",
    "Mental_Health_Score",
    "Physical_Health_Score",
    "Social_Isolation_Score",
    "Academic_Performance_Score"
]

risk_group_means = df.groupby("Academic_Failure_Risk")[important_columns].mean()
risk_group_means.columns = ["No Failure Risk", "Failure Risk"]

display(risk_group_means.round(2))
```

	No Failure Risk	Failure Risk
Daily_Social_Media_Hours	4.34	7.67
Daily_AI_Tool_Usage_Hours	2.53	3.51
Sleep_Hours	6.62	5.35
Physical_Activity_Hours	1.28	0.68
Mental_Health_Score	73.34	59.49
Physical_Health_Score	88.46	81.05
Social_Isolation_Score	4.24	5.37
Academic_Performance_Score	79.18	64.79

27. Create Useful Analysis Tables

These tables can be reused later in the project documentation or report.

```
In [27]: # Average academic performance by burnout level.
burnout_summary = df.groupby("Burnout_Level").agg(
    Students=("Student_ID", "count"),
    Average_Performance=("Academic_Performance_Score", "mean"),
    Average_Social_Media=("Daily_Social_Media_Hours", "mean"),
    Average_AI_Usage=("Daily_AI_Tool_Usage_Hours", "mean"),
    Average_Sleep=("Sleep_Hours", "mean"),
    Average_Mental_Health=("Mental_Health_Score", "mean")
).reindex(burnout_order)

display(burnout_summary.round(2))
```

	Students	Average_Performance	Average_Social_Media	Average_AI_Usage
Burnout_Level				
Low	9000	80.86	3.55	2.34
Moderate	4200	75.79	5.54	2.82
High	1350	72.44	6.80	3.20
Severe	450	68.64	8.14	3.38

28. Key Findings

The following cell automatically calculates a few important values that can be used when writing the project conclusions.

Important: Correlation shows association, not causation. A relationship between two variables does not prove that one variable causes the other.

```
In [28]: print("=" * 80)
print("KEY FINDINGS")
print("=" * 80)

social_corr = df["Daily_Social_Media_Hours"].corr(df["Academic_Performance_Score"])
ai_corr = df["Daily_AI_Tool_Usage_Hours"].corr(df["Academic_Performance_Score"])
sleep_corr = df["Sleep_Hours"].corr(df["Academic_Performance_Score"])
mental_corr = df["Mental_Health_Score"].corr(df["Academic_Performance_Score"])
risk_corr = risk_correlations.iloc[0]

print(f"\n1. Social media vs academic performance: {social_corr:.3f}")
print(f"2. AI usage vs academic performance: {ai_corr:.3f}")
print(f"3. Sleep vs academic performance: {sleep_corr:.3f}")
print(f"4. Mental health vs academic performance: {mental_corr:.3f}")
print(f"5. Strongest numerical correlation with failure risk: {risk_correlations

highest_burnout_performance = burnout_performance["mean"].idxmax()
lowest_burnout_performance = burnout_performance["mean"].idxmin()
```

```
print(f"6. Highest average academic performance by burnout level: {highest_burnout_level}")
print(f"7. Lowest average academic performance by burnout level: {lowest_burnout_level}")
print("\nRemember: these results describe patterns in this dataset and should not be interpreted as proof of causation.")
```

=====

KEY FINDINGS

=====

1. Social media vs academic performance: -0.394
2. AI usage vs academic performance: -0.205
3. Sleep vs academic performance: 0.292
4. Mental health vs academic performance: 0.204
5. Strongest numerical correlation with failure risk: Mental_Health_Score (-0.356)
6. Highest average academic performance by burnout level: Low
7. Lowest average academic performance by burnout level: Severe

Remember: these results describe patterns in this dataset and should not be interpreted as proof of causation.

29. Conclusion

What We Analyzed

This notebook examined the main relationships in the cleaned student dataset, including:

- Social media usage and academic performance.
- AI tool usage and academic performance.
- Sleep, physical activity, and health indicators.
- Burnout levels and academic outcomes.
- Demographic and education-level differences.
- Factors associated with academic failure risk.

Next Step

The next stage can use these findings to prepare the dataset for **machine learning**, where `Academic_Failure_Risk` can be used as the target variable for classification models.